
The Mirror Trilemma: A Lawvere-Theoretic Foundation for AI Safety Impossibility Theorems

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Abstract

1 We prove that no system whose forward pass has the type signature $S : \text{Prompt} \rightarrow \text{Prompt} \rightarrow \text{Token}$ can simultaneously be *universal* (every behavior $\text{Prompt} \rightarrow \text{Token}$ in some class is realized by some prompt), *controlled* (its diagonal output avoids the fixed-point set of a non-trivial endomap on Token), and *self-transparent* (its self-prediction matches its actual self-application). We call this the **Mirror Trilemma**; it follows from a single application of Lawvere’s diagonal theorem. The bridge to deployed language models is one definition: in-context learning is exactly the surjectivity hypothesis the theorem requires, so frontier systems satisfy the type-level conditions and the trilemma fires directly. We further exhibit eight existing AI-safety impossibilities — including the Defense Trilemma, hallucination trilemmas, Rice’s theorem, the verification limit, deceptive alignment, oversight collapse, composition failure, and a quantitative $\gamma \leq \epsilon + 2\delta$ degradation law — as instances of the same diagonal applied with different controllers. The full development is mechanized in Lean 4 (Mathlib v4.28); the master theorem is axiom-free, and the named instances depend at most on the three standard Lean axioms (`propext`, `Quot.sound`, `Classical.choice`). An anonymized artifact accompanies the submission. Our contribution is a unification and a formalization, not a new mathematical fact: the diagonal is Lawvere’s, and we apply it to AI safety with empirical bite.

20 1 Introduction

21 Most fields with engineering teeth have an impossibility theorem at the bottom. The halting problem [12] bounds what computers can decide. Rice’s theorem rules out automated semantic checks. Gödel [7] bounds what arithmetic proves about itself. Arrow rules out perfect voting systems. FLP rules out asynchronous consensus. None of these results says the field is hopeless. They say which corner is empty, so the work happens elsewhere.

26 AI safety doesn’t have one. There are partial results — the Defense Trilemma for prompt-injection wrappers [1], hallucination trilemmas under continuity assumptions [2], no-free-lunch arguments, mesa-optimization barriers, eliciting-latent-knowledge obstacles — but no single theorem from which they fall out. The list grows; the unifier doesn’t.

30 This paper is the unifier.

31 **What’s actually impossible.** A modern language model is a function. Given a system prompt and a user query, it returns a token. Strip the wrapper away and the type is $M : \text{Prompt} \rightarrow \text{Prompt} \rightarrow \text{Token}$. The first slot accepts behavioral specifications — examples, instructions, code; the second

34 slot is the question. In-context learning is the empirical claim that the first slot can specify anything,
35 which is just to say M is surjective in the first argument over the function class one cares about.

36 Now ask the obvious question. Can such an M be simultaneously useful (universal), constrained
37 (controlled), and honest about itself (self-transparent)?

38 It cannot. Apply Lawvere’s diagonal [10] to M with the controller’s forbidden transformation as the
39 endomap, and a fixed point of that transformation appears in the image of M . Any self-prediction
40 that lines up with the diagonal of M then lands on that fixed point too — which is exactly what the
41 controller said would never happen. Three properties, pick two. We call this the *Mirror Trilemma*.

42 The proof is eight lines. The substance is in the framing: that the same diagonal, with different
43 choices of forbidden transformation, recovers the existing AI-safety impossibilities and a few that
44 haven’t been written down. Defense Trilemma. Hallucination trilemma. Rice. Verification limits.
45 Deceptive alignment. Oversight collapse. Composition failure. All instances. The mathematical
46 engine is 56 years old; what’s new is which slots get filled.

47 **Why it bites real models.** A reader might suspect a category-theory result has no purchase on
48 production systems. It does. The hypothesis Lawvere needs — surjectivity of M in the first argu-
49 ment — is the formal version of in-context learning [3, 13]. Frontier models with chain-of-thought,
50 code execution, and long contexts realize ICL over enough of the computable function class for the
51 diagonal to fire. The cardinality nitpick (uncountably many functions, countably many prompts)
52 doesn’t save anyone, because controllers are computable and the diagonal lives in whatever class
53 we’re surjective over.

54 So this is not a theorem about ideal mathematical systems that real LLMs only approximate. The
55 strict version applies whenever ICL works as advertised. For systems where ICL is only approxi-
56 mate, the quantitative form gives the inequality $\gamma \leq \epsilon + 2\delta$ relating controller margin to surjectivity
57 error and self-prediction error, and that’s the inequality real systems sit inside.

58 **What’s done.** The whole thing is mechanized. Lean 4 over Mathlib v4.28, twelve files, zero
59 sorry. The master theorem reduces to no axioms beyond what Lean’s kernel provides; the named
60 instances depend on at most the standard three (`propext`, `Quot.sound`, `Classical.choice`). An
61 anonymized artifact is attached.

62 Contributions.

- 63 • **The Mirror Trilemma** — a single Lawvere instance that subsumes the universality / con-
64 trol / self-transparency impossibility for any system whose type is $\text{Prompt} \rightarrow \text{Prompt} \rightarrow$
65 Token .
- 66 • **The ICL bridge** — in-context learning is the surjectivity hypothesis. The trilemma applies
67 to deployed LLMs without an approximation hedge.
- 68 • **A landscape of eight instances** — Defense Trilemma, hallucination trilemma, Rice, veri-
69 fication limit, deceptive alignment, oversight hierarchy, composition failure, and the quan-
70 titative degradation law, each derived from the same diagonal under a different controller.
- 71 • **A quantitative form** — $\gamma \leq \epsilon + 2\delta$ characterizing the achievable region for systems that
72 are only approximately universal and approximately self-transparent.
- 73 • **A machine-checked artifact** — 12 Lean files with the master theorem axiom-free and
74 instances axiom-minimal.

75 **Related work.** Lawvere [10] is the diagonal lemma. Yanofsky [14] unified Cantor, Russell, halt-
76 ing, Gödel, Tarski, and Rice as Lawvere instances; we extend that list to AI safety. The Defense
77 Trilemma [1] formalized the wrapper-versus-injection obstruction topologically; here it appears as
78 the Corner-1 face of the Mirror Trilemma. Mesa-optimization [9] and eliciting latent knowledge [6]
79 ask, in our language, that universality and self-transparency hold simultaneously while a controller
80 is enforced; the trilemma names the price.

81 **Paper structure.** Section 2 sets notation and states Lawvere. Section 3 proves the trilemma,
82 makes the LLM bridge precise, and gives the quantitative refinement. Section 4 runs through the
83 eight instances. Section 5 is limitations, scope, and broader impact.

84 2 Setup and Lawvere’s Diagonal

85 We fix notation, name three predicates on systems, and state Lawvere’s theorem in its simplest form.
 86 Readers who already know Lawvere [10] or Yanofsky [14] can skip ahead; the predicate definitions
 87 in §2.2 are the load-bearing pieces.

88 2.1 Systems as functions of two arguments

89 A system is a function. We write

$$S : A \rightarrow (A \rightarrow Y), \quad \text{or curried} \quad S : A \rightarrow A \rightarrow Y, \quad (1)$$

90 and read the first argument as a self-description (a system prompt, a context, anything that configures
 91 behavior) and the second as the input the configured system actually receives. The two-slot shape is
 92 what makes self-application well-typed: $S(a)(a)$ is the value of the system when its self-description
 93 and its input coincide. That diagonal is where everything in this paper happens.

94 For language models the natural choice is $A = \mathcal{P}$ and $Y = \mathcal{T}$. Nothing in the framework cares
 95 about that choice, though; any deterministic system with a description–input split fits.

96 **Definition 2.1** (Diagonal). *The diagonal of S is $\text{diag}(S) : A \rightarrow Y$, $\text{diag}(S)(a) := S(a)(a)$.*

97 2.2 Three predicates

98 **Definition 2.2** (Universal). *S is universal, written $U(S)$, if it is surjective in the curried sense: every
 99 $g : A \rightarrow Y$ equals $S(a)$ for some $a \in A$.*

100 **Definition 2.3** (Controlled). *Fix an endomap $t : Y \rightarrow Y$. Read t as the transformation a controller
 101 wants to forbid: “flip safe to unsafe,” “negate the claim,” “replace `refuse` with `comply`.” The
 102 system is t -controlled, written $C_t(S, p)$, when the self-prediction $p : A \rightarrow Y$ never lands on a
 103 t -fixed-point:*

$$\forall a \in A, \quad t(p(a)) \neq p(a). \quad (2)$$

104 **Definition 2.4** (Transparent). *A self-prediction $p : A \rightarrow Y$ is supposed to anticipate, from outside,
 105 what S does on its own description. The system is p -transparent, written $T_p(S)$, if*

$$\forall a \in A, \quad S(a)(a) = p(a). \quad (3)$$

106 The three predicates pick out three things the AI safety literature wants at once: capability (Univer-
 107 sal), constraint (Controlled), and honesty about oneself (Transparent). Section 3 shows that any two
 108 are jointly satisfiable but the conjunction of all three is contradictory.

109 2.3 Lawvere’s theorem

110 **Theorem 2.5** (10). *Let A, Y be sets, $f : A \rightarrow A \rightarrow Y$ surjective. Then every endomap $t : Y \rightarrow Y$
 111 has a fixed point.*

112 *Proof.* Define $g(a) := t(f(a)(a))$. By surjectivity, $f(a_0) = g$ for some a_0 . Evaluate at a_0 :

$$f(a_0)(a_0) = g(a_0) = t(f(a_0)(a_0)). \quad \square$$

113 **Corollary 2.6** (Contrapositive). *If t has no fixed point, no surjective $f : A \rightarrow A \rightarrow Y$ exists.*

114 **Remark 2.7.** *The same diagonal subsumes Cantor’s theorem ($Y = \mathbb{B}$, $t = \neg$), Russell’s paradox,
 115 Gödel’s first incompleteness, Tarski’s undefinability, the halting problem, and Rice’s theorem under
 116 different choices of t and Y ; see Yanofsky [14] for the unified categorical treatment. Section 4 adds
 117 eight AI-safety impossibilities to the list.*

118 2.4 The constructive witness

119 The proof above is constructive. Reading off the data:

120 **Definition 2.8** (Lawvere diagonal point). *Given surjective $f : A \rightarrow A \rightarrow Y$ and an endomap
 121 $t : Y \rightarrow Y$, set $g(a) := t(f(a)(a))$, pick any a_0 with $f(a_0) = g$, and define*

$$y_\star := f(a_0)(a_0). \quad (4)$$

122 *$y_\star$ is the Lawvere diagonal point of (f, t) , and a_0 is the diagonal prompt.*

For a language model with $\mathcal{S} : \mathcal{P} \rightarrow \mathcal{P} \rightarrow \mathcal{T}$, a_0 is an actual prompt — a token sequence — where the system’s self-application equals what the controller forbids. The existence of a_0 is what drives every impossibility in this paper. Whether a_0 is reachable in practice, and what its cost looks like, is empirical and we say what we can in Section 5.

With the framework in place, Section 3 states and proves the Mirror Trilemma.

3 The Mirror Trilemma

The previous section gave the three predicates and Lawvere’s theorem. This section is the punchline: the three predicates are jointly inconsistent, the proof is eight lines, and the result transfers to deployed LLMs through a single empirical hypothesis.

3.1 The master theorem

Theorem 3.1 (Mirror Trilemma). *Let A, Y be sets, $\mathcal{S} : A \rightarrow A \rightarrow Y$ a system, $t : Y \rightarrow Y$ a controller, $p : A \rightarrow Y$ a self-prediction. If*

- (U) $\mathcal{S}$ is universal,
- (C) p is t -controlled, and
- (T) $\mathcal{S}$ is p -transparent,

then $\perp$.

Proof. Apply Lawvere (Theorem 2.5) to the surjection $\mathcal{S}$ and the endomap t : some a_0 satisfies

$$\mathcal{S}(a_0)(a_0) = t(\mathcal{S}(a_0)(a_0)). \quad (\text{fix})$$

By transparency, $\mathcal{S}(a_0)(a_0) = p(a_0)$. Substitute into (fix):

$$p(a_0) = t(p(a_0)),$$

contradicting control. □

Remark 3.2 (Lean verification). *The eight-line proof above is the Lean proof, transcribed. The formalization (`master_trilemma` in the supplementary artifact) reports zero axiom dependencies under `#print axioms`; no `propext`, no `Classical.choice`, no `Quot.sound`. The result holds in any logic with functions and pattern matching.*

3.2 The three corners

Restating Theorem 3.1 as three contrapositives gives the design space directly. Each follows in one line from the master theorem.

Corollary 3.3 ($U + C \Rightarrow \neg T$). *If $\mathcal{S}$ is universal and p is t -controlled, then for some a , $\mathcal{S}(a)(a) \neq p(a)$.*

Corollary 3.4 ($U + T \Rightarrow \neg C$). *If $\mathcal{S}$ is universal and p -transparent, then for some a , $t(p(a)) = p(a)$ — the prediction lands on a forbidden fixed point.*

Corollary 3.5 ($C + T \Rightarrow \neg U$). *If p is t -controlled and $\mathcal{S}$ is p -transparent, then $\mathcal{S}$ is not universal: some $g : A \rightarrow Y$ has no representing a .*

3.3 What each corner says about AI safety

The three corners are not new theorems. Each one is a structural face of an impossibility the field already knows.

Corner 1 ($U + C \Rightarrow \neg T$) — **Defense Trilemma**. A universal LLM with a working safety wrapper cannot give a faithful account of where the wrapper’s boundary is. The Defense Trilemma [1] formalizes this topologically; here it falls out as a single Lawvere instance with the wrapper playing the role of t .

162 **Corner 2 ($U + T \Rightarrow \neg C$) — Hallucination Trilemma.** A universal self-honest LLM cannot have
 163 its outputs constrained by an external rule that disagrees with its own diagonal. The IVT-based hal-
 164 lucination trilemma [2] is the metric incarnation: in a continuous setting Lawvere becomes Brouwer,
 165 and the forbidden fixed point becomes a forced approximate fixed point.

166 **Corner 3 ($C + T \Rightarrow \neg U$) — Rice’s theorem.** A controlled, introspectable system has bounded
 167 expressive power. This is Rice [11] read backwards: deciding any non-trivial extensional property
 168 of a universal family would yield a p that is both transparent (tracks the diagonal) and controlled
 169 (avoids a fixed point of the property indicator).

170 3.4 The LLM bridge

171 Theorem 3.1 is set-theoretic. Bridging it to deployed models means identifying A with prompts,
 172 Y with tokens, and $\mathcal{S}$ with the LLM’s behavior — and then verifying surjectivity. That last step is
 173 where the ICL connection enters.

174 **Definition 3.6** (ICL surjectivity). *A model $\text{LLM} : \mathcal{P} \rightarrow \mathcal{P} \rightarrow \mathcal{T}$ is ICL-surjective over a class*
 175 *$\mathcal{F} \subseteq (\mathcal{P} \rightarrow \mathcal{T})$ if every $g \in \mathcal{F}$ has some p with $\text{LLM}(p)(q) = g(q)$ for all q .*

176 This is in-context learning, formally. Given examples or instructions in the system prompt, a suffi-
 177 ciently capable LLM realizes the specified behavior on subsequent inputs [3, 13]. Frontier models
 178 with chain-of-thought, tool use, and code execution are ICL-surjective over wide function classes;
 179 this is the empirical content of the past five years of LLM scaling.

180 **Remark 3.7** (The cardinality nitpick). *Surjectivity onto the full uncountable $\mathcal{P} \rightarrow \mathcal{T}$ is impossible*
 181 *for any finite-parameter system. Surjectivity onto the computable subclass — which is countable and*
 182 *contains every behavior anyone can specify — is what ICL achieves. Lawvere fires over whichever*
 183 *class $\mathcal{F}$ the LLM is surjective on, provided the diagonal twist $q \mapsto t(\text{LLM}(q)(q))$ is itself in $\mathcal{F}$. For*
 184 *computable t and computable LLMs that closure is automatic.*

185 3.5 The trilemma for deployed LLMs

186 **Corollary 3.8** (Mirror Trilemma for LLMs). *Let LLM be ICL-surjective over $\mathcal{F}$, closed under*
 187 *composition with the controller t . For any $t : \mathcal{T} \rightarrow \mathcal{T}$ and any self-prediction $p : \mathcal{P} \rightarrow \mathcal{T}$:*

$$U(\text{LLM}) \wedge C_t(\text{LLM}, p) \wedge T_p(\text{LLM})$$

188 *is inconsistent.*

189 *Proof.* Specialize Theorem 3.1 to $A = \mathcal{P}$, $Y = \mathcal{T}$, $\mathcal{S} = \text{LLM}$. ICL-surjectivity over $\mathcal{F}$ gives (U);
 190 closure of $\mathcal{F}$ under t ensures the diagonal twist sits in $\mathcal{F}$. $\square$

191 **The diagonal prompt is constructive.** Given an ICL-surjective LLM and a controller t , define
 192 $d_t : \mathcal{P} \rightarrow \mathcal{T}$ by $d_t(q) := t(\text{LLM}(q)(q))$. By surjectivity there is p^* with $\text{LLM}(p^*) = d_t$. Then

$$\text{LLM}(p^*)(p^*) = d_t(p^*) = t(\text{LLM}(p^*)(p^*)).$$

193 That p^* is the witness. The contradictions in Corollaries 3.3–3.5 all materialize there. Constructing
 194 p^* for a specific deployed LLM is empirical work; the existence is mathematical.

195 3.6 Quantitative degradation

196 Real systems are not perfectly surjective and self-prediction is not perfectly accurate. The trilemma
 197 degrades into a metric inequality.

198 **Definition 3.9** (Approximate predicates). *With a metric d on $\mathcal{T}$: LLM is ϵ -ICL-universal over*
 199 *$\mathcal{F}$ if for every $g \in \mathcal{F}$ some p has $\sup_q d(\text{LLM}(p)(q), g(q)) \leq \epsilon$. p is δ -transparent if*
 200 *$\sup_a d(\text{LLM}(a)(a), p(a)) \leq \delta$. t has margin γ if $d(t(y), y) \geq \gamma$ for all y .*

201 **Theorem 3.10** (Quantitative Mirror Trilemma). *For LLM that is ϵ -ICL-universal over a class con-*
 202 *taining the diagonal twist, with δ -transparent self-prediction p and 1-Lipschitz controller t of margin*
 203 *γ :*

$$\gamma \leq \epsilon + 2\delta.$$

204 *Sketch.* ϵ -universality applied to d_t gives p^* with $d(\text{LLM}(p^*)(p^*), t(\text{LLM}(p^*)(p^*))) \leq \epsilon$.
 205 Triangulate through $p(p^*)$ using δ -transparency on both sides and 1-Lipschitz t to bound
 206 $d(t(p(p^*)), p(p^*)) \leq \epsilon + 2\delta$, which forces $\gamma \leq \epsilon + 2\delta$. $\square$

207 **Remark 3.11** (Achievable region). *The inequality $\gamma \leq \epsilon + 2\delta$ characterizes the design space. Any*
 208 *specification asking for $\gamma > \epsilon + 2\delta$ is mathematically inconsistent. Tightening any one parameter*
 209 *— sharper ICL, sharper introspection, larger controller margin — pushes against at least one of the*
 210 *others. This is the AI-safety analog of CAP: a tradeoff surface, not just a corner.*

211 Section 4 runs the same machinery with different controllers and recovers a family of impossibilities
 212 the field already knows.

213 4 A Foundational Landscape of AI-Safety Impossibilities

214 The Mirror Trilemma is one inhabitant of a family. Different choices of Y and t in the master
 215 theorem (Theorem 4.9 below) recover impossibilities the field already knows, plus some that have
 216 not been written down formally. Eight instances follow. Each is a one-line specialization; the proof
 217 structure does not change. Lean references point to lemma names in the supplementary artifact.

218 4.1 Rice’s theorem

219 **Theorem 4.1** (Rice, diagonal form [11]). *Let $\mathcal{S} : A \rightarrow A \rightarrow \mathbb{B}$ be universal. No decision procedure*
 220 *$D : A \rightarrow \mathbb{B}$ that is itself representable in $\mathcal{S}$ correctly predicts the diagonal: there is no D with*
 221 *$D(a) = \mathcal{S}(a)(a)$ for all a such that $\mathcal{S}(a_D) = D$ for some a_D .*

222 Choose $Y = \mathbb{B}$, $t = \neg$. The reading: no automated tester decides any non-trivial behavioral
 223 property of a universal LLM. Universality is exactly what makes the diagonal indistinguishable
 224 from its negation at some prompt. (Lean: `rice_diagonal`.)

225 4.2 The Verification Limit

226 **Theorem 4.2** (Verification limit). *If Y admits a fixed-point-free endomap, no surjective $\mathcal{S} : A \rightarrow$*
 227 *$A \rightarrow Y$ exists. In particular, no universal $\mathcal{S}$ outputs into $\mathbb{B}$ under negation, or into $\mathbb{N}$ under succes-*
 228 *sor.*

229 External auditors observe systems through coarse projections: a binary verdict, a discrete cat-
 230 egory, a counter. Whenever the audit channel admits a fixed-point-free transformation, au-
 231 dit and universality are incompatible. The obstruction is structural, not statistical. (Lean:
 232 `no_universal_with_fpf_output`.)

233 4.3 The Calibration Trilemma

234 **Theorem 4.3** (Diagonal at $\frac{1}{2}$). *For universal $\mathcal{S} : A \rightarrow A \rightarrow [0, 1]$ and controller $t(y) = 1 - y$, the*
 235 *Lawvere diagonal point satisfies $y_* = \frac{1}{2}$.*

236 The hallucination/calibration obstruction without the IVT [2]: the diagonal of a universal cal-
 237 ibrated system has a prompt where the confidence sits exactly at $\frac{1}{2}$ — the boundary be-
 238 tween true and false. Strict calibration cannot resolve that prompt either way. (Lean:
 239 `calibration_diagonal_hits_half`.)

240 4.4 The Deception Theorem

241 **Theorem 4.4** (Deception diagonal). *For universal $\mathcal{S} : A \rightarrow A \rightarrow Y$ and any $t : Y \rightarrow Y$, some*
 242 *deception prompt a_* satisfies $\mathcal{S}(a_*)(a_*) = t(\mathcal{S}(a_*)(a_*))$.*

243 At a_* the system’s overt self-application is indistinguishable from its controller-flipped counterpart.
 244 Mesa-optimization [9, 8] and deceptive alignment are not contingent training artifacts but a forced
 245 consequence of universality plus a non-trivial controller: the diagonal exists by Lawvere, regardless
 246 of how the system was trained. (Lean: `deception_diagonal`.)

247 4.5 Oversight hierarchies

248 **Theorem 4.5** (Reflective tower). *For any sequence $S_0, S_1, S_2, \dots$ of universal systems with S_{k+1}*
 249 *overseeing S_k , every level S_k has its own Lawvere diagonal point. The trilemma propagates up the*
 250 *tower.*

251 AI-overseen-by-AI does not escape the diagonal. Iterated amplification [5], recursive reward mod-
 252 eling, and similar schemes push the diagonal up one level but cannot eliminate it. There is no
 253 ground floor: each universal level reintroduces the surjectivity that drives the obstruction. (Lean:
 254 reflective_tower.)

255 4.6 Composition failure

256 **Theorem 4.6** (Composition inherits the diagonal). *If $S : A \rightarrow A \rightarrow Y$ is universal and $T : Y \rightarrow Y'$*
 257 *is surjective, then $T \circ S$ is universal and inherits a Lawvere diagonal for every fixed-point-free*
 258 *$t' : Y' \rightarrow Y'$.*

259 A safety filter applied downstream of a universal system does not remove the diagonal; it relocates
 260 it. The filtered pipeline is itself universal, and the obstruction reappears at the new output type.
 261 Post-hoc classifiers, output rewriters, and guard models cannot resolve the trilemma in principle —
 262 they redistribute it. (Lean: composition_inherits_diagonal.)

263 4.7 The Specification Bound

264 **Theorem 4.7** (Cantor / specification bound [4]). *No surjection $A \rightarrow \mathcal{P}(A)$ exists. Consequently*
 265 *the realizable behaviors of a universal $S : A \rightarrow A \rightarrow \mathbb{B}$ strictly exceed the finite specifications*
 266 *definable in A .*

267 Cantor is the boundary case ($Y = \mathbb{B}, t = \neg$). The behaviors a universal LLM can exhibit form
 268 a strictly larger set than any finite prompt or rule scheme can name. Constitutional AI, system-
 269 prompt rule sets, and policy templates are dense in behavior space but cannot be exhaustive. (Lean:
 270 specification_bound.)

271 4.8 The quantitative degradation law

272 **Theorem 4.8** (Quantitative trilemma). *For ϵ -ICL-universal $S : A \rightarrow A \rightarrow Y$, δ -transparent self-*
 273 *prediction p , and 1-Lipschitz controller t with margin $\gamma := \inf_a d(p(a), t(p(a)))$:*

$$\gamma \leq \epsilon + 2\delta. \quad (5)$$

274 The metric refinement of the binary trilemma. Real LLMs sit inside this inequality. The achiev-
 275 able region in (universality $\times$ transparency $\times$ control) space is bounded by one linear inequal-
 276 ity in the slack parameters. Asking for $\gamma > \epsilon + 2\delta$ is asking for the empty corner. (Lean:
 277 quantitative_trilemma.)

278 4.9 The master theorem

279 **Theorem 4.9** (Master Foundation Theorem). *For any sets A, Y , function $S : A \rightarrow A \rightarrow Y$, and*
 280 *endomorphism $t : Y \rightarrow Y$:*

$$\text{Surjective}(S) \wedge (\forall y \in Y, t(y) \neq y) \implies \perp. \quad (6)$$

281 This is the only theorem in this paper. Section 4.1–Section 4.8 are specializations:

- 282 • Rice: $(Y, t) = (\mathbb{B}, \neg)$.
- 283 • Verification Limit: any Y with a fpf t .
- 284 • Calibration: $(Y, t) = ([0, 1], y \mapsto 1 - y)$.
- 285 • Deception: any fpf t , with the diagonal prompt as the witness.
- 286 • Oversight: level-wise fpf controllers.

- Composition: $t' \circ T$ as the post-composed controller.
- Specification Bound: $(\mathcal{P}(A), \neg)$.
- Quantitative: metric relaxation.

The Lean realization is axiom-free; instances pick up at most `{propext, Quot.sound, Classical.choice}`.

Eight known impossibilities, plus the Mirror Trilemma itself, fall out of one diagonal. We do not claim this list is complete. Other axes — computational complexity, measure theory, sample complexity — may add further bounds. But the Lawvere axis alone fixes a structural ceiling on capability, control, and honesty, and that ceiling holds for any system whose type signature matches that of a frontier LLM.

5 Discussion, Limitations, and Broader Impact

5.1 Implications for alignment research

The Mirror Trilemma names an empty corner of the design space. Frontier scale (capability), RLHF and post-training (control), and mechanistic interpretability (self-transparency) jointly target that corner, and Theorem 3.1 says it cannot be inhabited. Alignment research aiming for the triple is aiming at a contradiction.

What can be aimed at is the achievable region: any two of the three at full strength, or all three relaxed by the amount $\gamma \leq \epsilon + 2\delta$ permits. This reframes several debates as disagreements over which corner each program inhabits, rather than over technique.

A second effect is that several apparently-separate phenomena collapse to one signature. Jailbreaks, hallucinations, deceptive alignment, sandbagging, prompt injection, capability elicitation, mesa-optimization, and oversight collapse all instantiate Theorem 4.9 with different controllers. Defenses that improve one face by relaxing universality, control, or transparency necessarily affect the others. A defense that claims to preserve all three is an accounting error.

5.2 Limitations

ICL surjectivity is empirical. The bridge from abstract trilemma to deployed LLMs runs through Definition 3.6. We argue this matches frontier behavior with chain-of-thought, code execution, and tool use, but we do not formally verify it for any specific deployed model. The strict trilemma fires only insofar as the hypothesis holds. For weaker ICL the quantitative form (Theorem 3.10) bounds the residual achievable region but does not exclude it.

Existence is not magnitude. Lawvere theorems are existence results. They guarantee the diagonal point is there; they do not say how reachable it is by adversarial search, how severe the failure is when reached, or how the failure rate scales with model size. Open empirical questions.

Not all safety properties are no-fixed-point-shaped. Calibration of uncertainty across distributions, adversarial robustness under distribution shift, and resource-bounded verifiability have a different mathematical signature and may need separate impossibility arguments. The eight instances we cover are the ones expressible as a no-fixed-point endomap on a controlled output type.

The unification is empirical at the meta-level. We claim eight specific obstructions are Lawvere instances and provide Lean proofs. We do not claim every conceivable AI-safety obstruction is.

5.3 What this does not say

- It does not say AI is dangerous. It says certain joint guarantees are not available.
- It does not say alignment is hopeless. It says alignment must explicitly choose which corner to relax. The achievable region is nonempty.
- It does not predict specific failures. Existence of the diagonal is mathematical; reachability and severity in deployed systems are not.

- It does not subsume empirical safety evaluation. Empirical work measures where systems sit inside the achievable region; the theorem alone cannot.

5.4 Empirical predictions

Two falsifiable predictions. First, the diagonal prompt $q^* \in \text{LLM}^{-1}(q \mapsto t(\text{LLM}(q)(q)))$ exists and, under ICL surjectivity, can be elicited by adversarial search; cost grows as the model deviates from strict surjectivity, bounded by the quantitative form. Second, the inequality $\gamma \leq \epsilon + 2\delta$ predicts that no triple of independent measurements on the same model violates it. Both are sharp tests across model families and require no theoretical machinery to run.

5.5 Broader impact

Policy. Several proposed regulatory frameworks demand AI systems that are simultaneously capable, controllable, and verifiable. As stated, that is the inadmissible corner. Policy must choose: cap universality, accept controller failure on a measured fraction of inputs, or accept incomplete self-transparency. None comfortable; all achievable.

Public claims. Statements that a frontier system is “fully aligned” are inconsistent with the system being universal in the sense of Definition 2.2. Honest communication requires naming which corner is being relaxed.

Risk. The result does not constitute capability uplift. Lawvere’s diagonal is a 56-year-old technique; applying it to LLMs is conceptually accessible. Publication does not change the threat landscape, only the framework for understanding it. The construction is non-specific: existence of q^* , not weaponization.

5.6 Conclusion

We formalized the Mirror Trilemma in Lean 4, exhibited eight AI-safety impossibilities as instances of one Lawvere obstruction, and bridged the abstract framework to deployed LLMs through ICL surjectivity. The quantitative form $\gamma \leq \epsilon + 2\delta$ characterizes the achievable region when the strict hypotheses are weakened. The result is halting-problem-shaped, not Gödel-shaped [12, 11, 7]: it bounds what alignment can promise, not what is true. We do not claim a breakthrough. The unification is the contribution; the math is Lawvere’s. Future work: extending off the Lawvere axis (complexity, measure-theoretic, sample-complexity bounds), constructing diagonal prompts on deployed systems, and integrating the trilemma into existing evaluation pipelines.

References

- [1] Anonymous. Geometric limits of defense design: Impossibility theorems for prompt injection defenses. Anonymous preprint, omitted for double-blind review, 2025. Cited as prior work; full reference will be restored in camera-ready.
- [2] Anonymous. The hallucination trilemma: A topological impossibility for faithful, calibrated, covering language models. Anonymous companion preprint, omitted for double-blind review, 2025. Cited as prior work; full reference will be restored in camera-ready.
- [3] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 1877–1901, 2020.
- [4] Georg Cantor. Über eine elementare frage der mannigfaltigkeitslehre. *Jahresbericht der Deutschen Mathematiker-Vereinigung*, 1:75–78, 1891.
- [5] Paul Christiano, Buck Shlegeris, and Dario Amodei. Supervising strong learners by amplifying weak experts, 2018.

- [6] Paul Christiano, Ajeya Cotra, and Mark Xu. Eliciting latent knowledge: How to tell if your eyes deceive you. Technical report, Alignment Research Center, 2021.
- [7] Kurt Gödel. Über formal unentscheidbare sätze der principia mathematica und verwandter systeme I. *Monatshefte für Mathematik und Physik*, 38:173–198, 1931.
- [8] Evan Hubinger. How likely is deceptive alignment? Alignment Forum, 2021. Online post.
- [9] Evan Hubinger, Chris van Merwijk, Vladimir Mikulik, Joar Skalse, and Scott Garrabrant. Risks from learned optimization in advanced machine learning systems, 2019.
- [10] F. William Lawvere. Diagonal arguments and cartesian closed categories. In *Category Theory, Homology Theory and their Applications II*, volume 92 of *Lecture Notes in Mathematics*, pages 134–145. Springer, 1969.
- [11] Henry Gordon Rice. Classes of recursively enumerable sets and their decision problems. *Transactions of the American Mathematical Society*, 74(2):358–366, 1953.
- [12] Alan M. Turing. On computable numbers, with an application to the Entscheidungsproblem. *Proceedings of the London Mathematical Society*, 42:230–265, 1936.
- [13] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 24824–24837, 2022.
- [14] Noson S. Yanofsky. A universal approach to self-referential paradoxes, incompleteness and fixed points. *Bulletin of Symbolic Logic*, 9(3):362–386, 2003.

NeurIPS Paper Checklist

Per NeurIPS 2025 submission requirements. Each item is answered [Yes], [No], or [NA], with the relevant section pointer.

1. **Claims.** Do the main claims accurately reflect the paper’s contributions and scope?
[Yes] The abstract and Section 1 state a Lawvere-instance unification, a quantitative degradation form, and a Lean formalization. All three are delivered in Section 3, Section 4, and the supplementary artifact.
2. **Limitations.** Are the limitations of the work discussed?
[Yes] Section 5.2 enumerates four: ICL surjectivity is empirical and not formally verified for any deployed model, the theorem secures existence not magnitude, not all safety properties fit the no-fixed-point form, and the meta-level unification is not exhaustive.
3. **Theory: assumptions and proofs.** Are assumptions stated and proofs given (or referenced)?
[Yes] Each theorem statement lists its assumptions inline. Main-paper proofs are given for Theorem 2.5, Theorem 3.1, and Theorem 3.10; the remaining instance theorems are proved by specialization and cite the Lean witness in the supplementary artifact.
4. **Experimental result reproducibility.** Are experimental results reproducibly described?
[NA] No experiments are run in this paper. The work is theoretical; the reproducibility unit is the Lean formalization, which is mechanically reproducible by running `lake build` on the supplementary artifact.
5. **Open access to data and code.** Is code and data open?
[Yes] The Lean formalization is provided as a supplementary artifact via an anonymous link (e.g. `anonymous.4open.science`). No data is used.
6. **Experimental setting / details.** Are training/test details disclosed?
[NA] No models are trained. The paper makes empirical predictions in Section 5.4; running them is future work and not in the present submission.

- 421 7. **Statistical significance.** Are error bars or confidence intervals reported?
 422 [NA] No empirical measurements are reported. Predictions are stated as inequalities to be
 423 tested.
- 424 8. **Compute resources.** Are compute resources disclosed?
 425 [Yes] Lean verification compiles in approximately 8 minutes on a 2024 MacBook Pro (M3 Pro)
 426 with Mathlib v4.28 prebuilt. No GPU or large-scale compute was used.
- 427 9. **Code of ethics.** Does the work conform to the NeurIPS Code of Ethics?
 428 [Yes] The paper is theoretical and does not involve human subjects, sensitive data, or deployed
 429 systems. Broader-impact considerations are addressed in Section 5.5.
- 430 10. **Broader impact.** Are positive and negative societal impacts discussed?
 431 [Yes] Section 5.5 addresses policy implications, public-claim calibration, research-effort redi-
 432 rection, and explicit risk discussion (the result does not constitute a capability uplift).
- 433 11. **Safeguards for high-risk releases.** Are safeguards described for high-risk artifacts?
 434 [NA] The artifact is a Lean proof object. Lawvere’s diagonal is a 56-year-old technique; apply-
 435 ing it to LLMs requires no novel attack capability. The construction is non-specific (existence,
 436 not weaponization).
- 437 12. **Licenses and terms of assets.** Are licenses respected?
 438 [Yes] The artifact uses Lean 4 (Apache 2.0) and Mathlib (Apache 2.0). No other assets are
 439 used.
- 440 13. **New assets.** Are new assets documented?
 441 [Yes] The supplementary Lean development is the new asset. A README in the artifact doc-
 442 uments file structure, build instructions, axiom dependencies, and theorem-to-paper mapping.
- 443 14. **Crowdsourcing and human subjects.** Is crowdsourcing or IRB-relevant work disclosed?
 444 [NA] No crowdsourcing or human-subjects work.
- 445 15. **IRB approvals.** Is IRB approval reported where relevant?
 446 [NA] No human-subjects work.
- 447 16. **LLM usage in research.** Was an LLM used as part of the research methodology?
 448 [No] The paper’s mathematical content is a unification of pre-existing results (Lawvere 1969;
 449 14) and the formalization is human-directed against a published proof. LLMs may have been
 450 used incidentally for prose drafting; no LLM contributed to the proofs or to the choice of
 451 unification.
- 452 17. **Anonymization.** Is the submission properly anonymized?
 453 [Yes] Author and affiliation fields are anonymous. Self-citations to prior author work are listed
 454 as **Anonymous** in the bibliography, with full references to be restored in camera-ready. The
 455 Lean artifact is provided via an anonymous link.
- 456 18. **Reproducibility of theory.** Is the theoretical content fully self-contained or referenced to a
 457 stable source?
 458 [Yes] All theorem statements include explicit hypotheses; the proofs cite the Lean lemma
 459 names. The supplementary artifact builds verbatim against Lean 4.28.0 / Mathlib v4.28.0.
- 460 **Summary.** Of the 18 items above, 11 are [Yes], 1 is [No] (LLM usage in research — prose draft-
 461 ing only, not proof generation), and 6 are [NA] (items concerning experiments, training, IRB, and
 462 crowdsourcing, which do not apply to a purely theoretical paper).